Milestone 3 Report

Bayesian Reasoning for LLM Agents under Uncertain and Conflicting Evidence

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1. Problem Statement

LLM agents often use extra information while answering a question, such as a retrieved passage, an image caption or a tool output. This information is not always good. Some of it is wrong, some is irrelevant, and sometimes two pieces disagree. A good agent should become more sure when reliable evidence agrees with its answer and less sure when reliable evidence disagrees. If the evidence is irrelevant, its confidence should stay about the same.

BayesAgent [1] adds hidden variables and Bayesian inference to LLM reasoning and gives better-calibrated confidence on ScienceQA. But it does not study each piece of evidence separately, and it does not study how belief changes when evidence arrives one piece at a time.

Research question: How can BayesAgent be extended to handle several pieces of evidence with different reliability, and how should the agent’s belief and confidence change when new or conflicting evidence arrives?

M1 set the direction and M2 built the technical base. M3 asks what the evidence says, so the base experiments were checked, repeated properly and measured with uncertainty.

2. What I Did in M3

Audit of the M2 outputs. Before extending anything, I checked every saved output file. This found three silent bugs in my own runner:

- Prompts were cut to 2048 tokens. The end of the prompt, which holds the question, caption and retrieved knowledge, was lost for some questions.
- Answers were capped at 300 tokens, so many vPGM answer lists were never finished.
- Generation did not stop at the end marker `</EOR>`.

The input cut hit all 50 prompts of vPGM-N2 (the shortest was 2488 tokens), 12 of 50 of vPGM-N4 and 4 of 50 for vPGM-N3, Chameleon and Chameleon+. CoT was not affected: its longest prompt is 869 tokens and it never reached the output cap.

For vPGM-N2 the real question was removed entirely. The word “Hematite”, from the last in-context example, appears in all 150 saved responses and in none of the other files. The parser then fell back to a uniform 0.25 per option, which is where the 28% accuracy came from. For vPGM-N4 the 300-token cap left 90 of 150 answer lists incomplete and 7 questions fully uniform.

Fix and check. I replaced the generation code with one that has no input cut, a real stop on `</EOR>`, a length check and fixed seeds. I tested it on two failed N4 questions before using it:

both finished by themselves in 392 to 447 tokens with complete answer lists. I then re-ran the other four methods (Chameleon, Chameleon+, vPGM-N3 and vPGM-N4) on the same 50 questions with 600 new tokens. The M2 numbers for Chameleon (82%, ECE 11.19), Chameleon+ (82%, ECE 9.64), vPGM-N3 (90%, ECE 9.79) and vPGM-N4 (80%, ECE 7.47) are replaced by the numbers in this report. The first vPGM-N2 result is invalid and is not used anywhere.

Evaluation. On the fixed runs I measured accuracy, ECE, Brier score, log-loss and AUROC, each with a 95% interval, plus reliability diagrams, paired tests, a noise floor for ECE, and robustness to sample size and binning. I ran the BayesVPGM λ step with cross-validation, checked what the hidden variables do, and analysed the wrong answers. I also implemented the extension's update rule as code and checked its proved properties.

Done	Not done in M3
CoT, Chameleon, Chameleon+, vPGM-N3 and vPGM-N4 on the same 50 questions with a fixed runner	vPGM-N2 with the fixed runner (needs a full 50-question run with its long prompts)
Accuracy, ECE, Brier, log-loss, AUROC with 95% intervals, paired tests, noise floor	LLM experiments for ΔC (reliability and evidence distribution from the LLM)
Robustness: sample size, number of bins, effect of the runner bugs	More than one seed per method
BayesVPGM step with cross-validation, hidden-variable check	A-OKVQA clean and noisy data
Error analysis; update rule as code with checks	The full 2563-question benchmark

3. The Model, in Short (from M2)

Base model (vPGM). The LLM first makes a few small judgements, the hidden variables Z, then answers. In vPGM-N3, Z1 is whether outside knowledge is needed, Z2 whether the image caption can be trusted, and Z3 whether the detected text can be trusted. The official structure is Q,C,M,I,K → Z1; I → Z2; I → Z3; Q,Z1,Z2,Z3 → Y. The LLM states each probability in words, and

$$P(Y \mid X) \approx \Sigma_Z \, P(Y \mid Q, Z) \cdot P(Z \mid X),$$

with M = 3 sampled answers averaged.

BayesVPGM step. Let p_k be the vPGM probability of option k and n_k its number of votes. The final probability is $\pi_k = (n_k + \lambda \, p_k) / (M + \lambda)$. A small λ trusts the votes and a large λ trusts the vPGM probabilities. The repository's own posterior code writes the prior as p_k/τ, so $\tau = 1/\lambda$. The repository contains this posterior model and the loss classes, but no script that fits it on saved responses.

My extension. Every evidence item E_i gets a hidden reliability R_i in [0, 1], where $R_i = 0$ means ignore. The belief is a set of pseudo-counts α. A new item with reliability r and supported answer distribution $\bar{s}$ gives $\alpha' = \alpha + r \cdot \kappa \cdot \bar{s}$. In M2 I proved $\pi' = (1 - w)\pi + w \cdot \bar{s}$ with $w = r\kappa / (A + r\kappa)$, where A is the total of α. So r = 0 changes nothing, and one misleading item moves the belief by a bounded amount. The confidence change is $\Delta C = C_{after} - C_{before}$, measured on the answer that was most likely before the evidence.

Four predictions fixed in M2: reliable supporting evidence gives ΔC > 0; reliable contradicting evidence gives ΔC < 0; irrelevant evidence gives ΔC close to 0; misleading evidence moves a reliability-aware update less than an update that ignores reliability.

4. Experimental Setup and Metrics

Setup. All methods use the same 50 ScienceQA [4] questions (test_50: 21 with three options, 29 with four), the same model (Meta-Llama-3-8B-Instruct), the same prompts from the official code of [1], and the same metric code. Inference runs through Hugging Face transformers with GPU and CPU offloading, because the server driver (460, CUDA 11.2) is too old for vLLM. Sampling uses temperature 0.5, top-p 0.95 and at most 600 new tokens. CoT [2] and Chameleon [6] use one answer per question. Chameleon+ and vPGM use three, so Chameleon+ is the fair baseline for vPGM. Each method was run once, with seed 0. The ECE code is a standard 10-bin ECE on the top-answer confidence.

KPIs.

- Accuracy: share of questions answered correctly.
- ECE: 10 uniform bins, bin-size-weighted gap between accuracy and confidence; lower is better; shown $\times 10^2$.
- Brier score: squared error of the probabilities over all options; lower is better.
- Log-loss: like Brier but punishes confident wrong answers very hard; lower is better.
- AUROC of confidence: how often a correct answer has higher confidence than a wrong one (0.5 = no information, 1.0 = perfect).

CoT and Chameleon give only the confidence of the chosen answer. For their Brier score and log-loss I split the remaining probability equally over the other options. This is my assumption and affects only those two numbers.

Uncertainty. Accuracy has a Wilson 95% interval. The other metrics have a bootstrap 95% interval (10,000 resamples of the 50 questions). Methods are compared with paired tests on the same questions: McNemar for accuracy and Wilcoxon on the per-question Brier score.

Noise floor. Even a perfectly calibrated model has ECE above zero on 50 questions. I simulated a perfectly calibrated model 5000 times, using each method’s own confidences. If a measured ECE is not clearly above this floor, I cannot call the model miscalibrated.

5. Results: Baselines Against vPGM

Method	Answers / question	Correct /50	Accuracy % [95%]	ECE ×10² [95%]	ECE noise floor (mean / 95th)	Brier [95%]	Log-loss [95%]	AUROC
CoT	1	43	86.0 [74, 93]	13.35 [5.6, 23.4]	5.9 / 9.1	0.265 [0.12, 0.44]	0.549 [0.25, 0.91]	0.75
Chameleon	1	46	92.0 [81, 97]	8.08 [2.0, 16.7]	3.8 / 6.4	0.172 [0.05, 0.32]	0.353 [0.11, 0.65]	0.80
Chameleon+	3	46	92.0 [81, 97]	5.62 [1.1, 13.6]	2.7 / 5.6	0.142 [0.04, 0.27]	0.483 [0.07, 1.05]	0.70
vPGM-N3	3	44	88.0 [76, 94]	9.99 [8.5, 18.9]	9.7 / 14.2	0.171 [0.08, 0.29]	0.336 [0.19, 0.51]	0.94
vPGM-N4	3	44	88.0 [76, 94]	13.86 [9.7, 20.2]	9.2 / 13.5	0.146 [0.07, 0.24]	0.293 [0.17, 0.44]	0.95
vPGM-N2	3	not reported (invalid first run, see Section 8.4)						

Table 1. Main results on 50 ScienceQA questions with the fixed runner, one run per method.

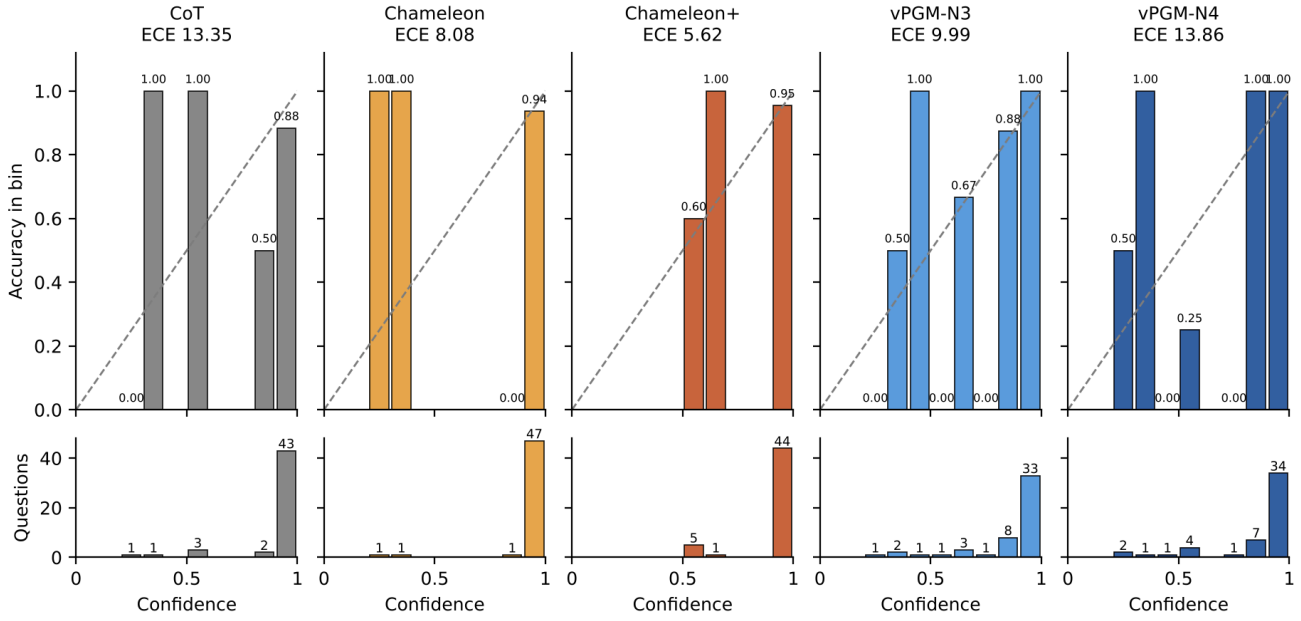


Figure 1. Reliability diagrams, 10 bins, N = 50.

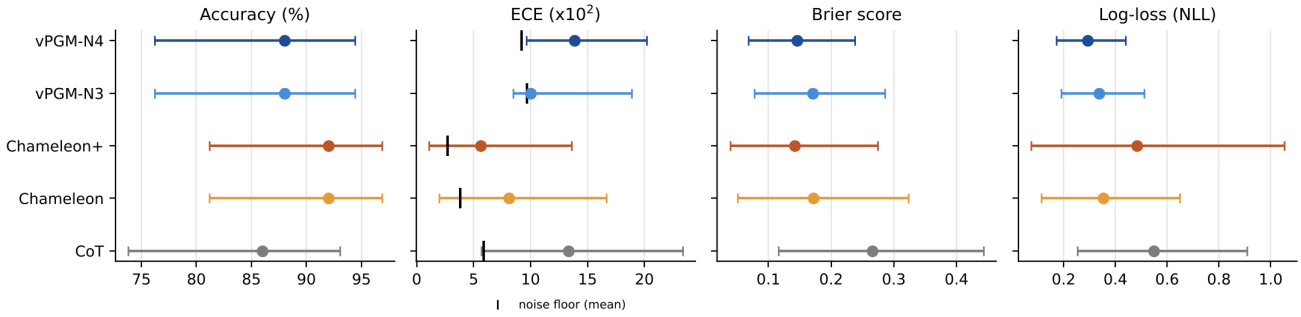


Figure 2. The five methods on four measures with 95% intervals and noise-floor marks.

5.1 Comparison with the paper

Method	Paper Acc.	Paper ECE	Mine Acc.	Mine ECE	What I can say
CoT	84.63	8.96	86.0	13.35	Accuracy similar; my ECE is higher and above its noise floor
Chameleon	85.29	9.62	92.0	8.08	Accuracy similar; ECE similar
Chameleon+	85.17	8.65	92.0	5.62	Accuracy similar; ECE lower but inside its noise floor
vPGM-N2	85.49	2.31	not reported	not reported	Invalid first run
vPGM-N3	86.38	1.67	88.0	9.99	Paper’s ECE not reproduced
vPGM-N4	86.54	2.15	88.0	13.86	Paper’s ECE not reproduced
BayesVPGM N3	86.38	1.05	87.6	10.07	Not reproduced (cross-validated, my script)
BayesVPGM N4	86.54	1.50	84.0	8.82	Not reproduced

Table 2. The paper (Table 1 of [1], 2563 questions) against my runs (50 questions).

Interpretation.

- **Accuracy (expected).** The paper’s accuracies (84.6 to 86.5) all lie inside my 95% intervals, so my setup is consistent with the paper. With 50 questions the interval is about ±9 points, so I cannot say more.
- **ECE (not what I hoped).** I did not see the paper’s large ECE advantage for vPGM, for two reasons visible in the data. First, the sample is small: the noise floor for vPGM-N3 and N4 is about 9 to 10, so even a perfectly calibrated vPGM would not show an ECE near 1.7 on 50 questions. Second, my vPGM is underconfident at the top. For N4, 41 of 50 answers have confidence 0.8 or higher and all 41 are right, so it says about 0.92 and is right 100% of the time. Being too careful adds to ECE just as being too sure does.
- **What this means.** This does not contradict the paper. It shows that a 50-question ECE cannot decide which method is better calibrated.

6. Are the Differences Real? (Statistical Tests)

Comparison (A − B)	Only A right / only B right	McNemar p	Acc. diff, points [95%]	ECE diff ×10² [95%]	Brier diff [95%]	Wilcoxon p (Brier)
vPGM-N3 − CoT	4 / 3	1.000	+2.0 [-8, 12]	-3.36 [-9.4, 7.6]	-0.094 [-0.250, +0.041]	0.144
vPGM-N4 − CoT	3 / 2	1.000	+2.0 [-6, 10]	+0.50 [-8.2, 8.3]	-0.119 [-0.257, -0.004]	0.216
vPGM-N3 − Chameleon+	2 / 4	0.688	-4.0 [-14, 6]	+4.37 [0.3, 12.7]	+0.029 [-0.088, +0.133]	<0.001
vPGM-N4 − Chameleon+	2 / 4	0.688	-4.0 [-14, 6]	+8.24 [-0.6, 15.5]	+0.004 [-0.111, +0.101]	<0.001
vPGM-N3 − Chameleon	1 / 3	0.625	-4.0 [-12, 4]	+1.91 [-2.6, 11.7]	-0.001 [-0.067, +0.059]	<0.001
vPGM-N4 − vPGM-N3	1 / 1	1.000	+0.0 [-6, 6]	+3.87 [-5.2, 7.8]	-0.025 [-0.082, +0.027]	0.097
Chameleon+ − Chameleon	1 / 1	1.000	+0.0 [-6, 6]	-2.46 [-10.3, 6.1]	-0.030 [-0.143, +0.094]	<0.001
Chameleon+ − CoT	3 / 0	0.250	+6.0 [0, 14]	-7.73 [-15.8, 1.5]	-0.123 [-0.249, -0.020]	<0.001

Table 3. Paired comparisons on the same 50 questions. With 8 comparisons, a p-value would need to be below about 0.006 to count as significant overall.

- **Accuracy.** In every comparison the two methods differ on only 2 to 7 of the 50 questions. The biggest gap (Chameleon+ over CoT, +6 points) is 3 questions to 0 with $p = 0.250$. I make no claim that one method is more accurate.
- **ECE.** All ECE intervals include zero except one: vPGM-N3 has a higher ECE than Chameleon+ (+4.37, [0.3, 12.7]), the opposite of the paper. Given the noise in ECE at this size, I do not read it as proof that Chameleon+ is better calibrated.
- **Brier.** For vPGM-N3 against Chameleon+ the Wilcoxon test is significant ($p < 0.001$) because Chameleon+ has the lower Brier score on 45 of 50 questions. But the mean difference is +0.029 and its interval includes zero. The Chameleon methods say 0.95 or more on almost everything, so they are nearly perfect when right. vPGM spreads some probability over the other options and pays a small price on every correct answer, but loses much less on the few wrong ones. Both statements are true and neither is a clear win.

7. Probability Quality: Does the Confidence Mean Something?

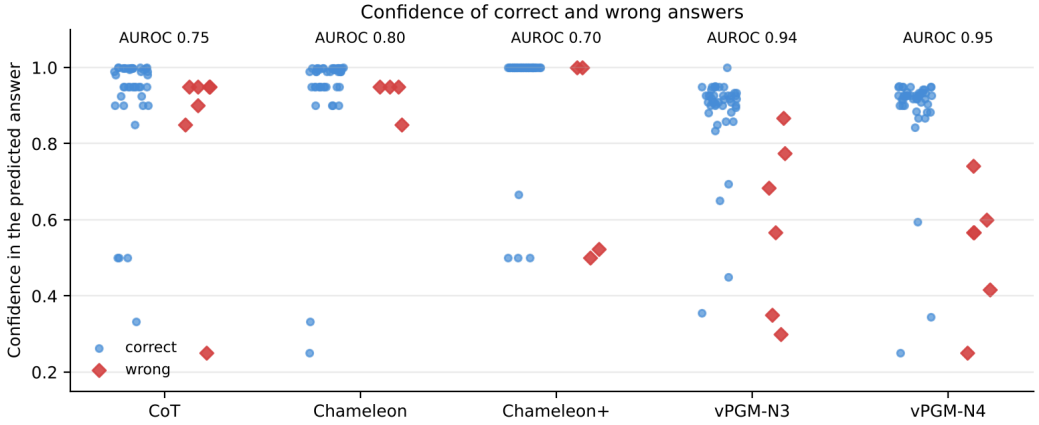


Figure 3. Confidence of correct (blue) and wrong (red) answers, AUROC above each method.

- CoT and Chameleon give almost the same confidence to right and wrong answers: wrong 0.83 and 0.92, right 0.92 and 0.95.
- Chameleon+ is better (0.76 against 0.96), but two of its four wrong answers still have confidence 1.00.
- The vPGM variants give wrong answers an average confidence of 0.59 (N3) and 0.52 (N4), clearly below their right answers (0.88). This is what AUROC measures: 0.94 and 0.95 for vPGM against 0.70 to 0.80 for the baselines.

Is the gap real? Comparing AUROC on the same resampled questions, the difference to CoT is +0.19 [0.04, 0.35] for N3 and +0.20 [0.08, 0.35] for N4, so the interval stays above zero. Against Chameleon and Chameleon+ the intervals just reach zero (for example [−0.04, 0.29] for N3 against Chameleon). Each AUROC rests on only 4 to 7 wrong answers, so the intervals are wide.

Conclusion: vPGM's better ranking of its own mistakes is supported against CoT and suggestive against the Chameleon methods.

Why the measures disagree. vPGM-N4 has the highest ECE (13.86) because its confident answers are always right, but the best log-loss (0.293) because it never gives a wrong answer a confidence near 1. Chameleon+ has the lowest ECE and Brier score, but its log-loss (0.483) is worse than both vPGM variants because two wrong answers at confidence 1.00 cost a lot. ECE rewards being right on average and log-loss punishes confident mistakes. For an agent that must know when not to trust itself, the second matters more, so I do not use ECE alone.

8. Robustness

8.1 Sample size

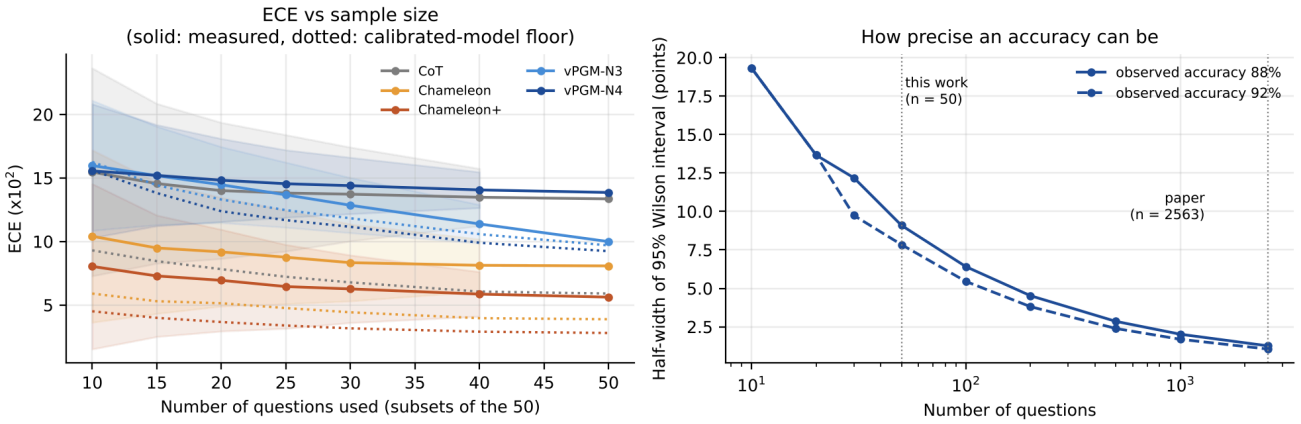


Figure 4. ECE against sample size with the noise floor (left); accuracy interval width against n (right).

ECE falls as more questions are used, and so does the noise floor, so ECE on a small sample is too high. vPGM-N3 sits on its own floor at every size up to 50, so it cannot be told apart from a perfectly calibrated model. vPGM-N4 stays about 4 to 5 points above its average floor, but only just above the 95th percentile of the floor at 50 questions. The Chameleon methods are about 3 to 4 points above their floor. An accuracy interval is about ±9 points at n = 50 and about ±1.3 at n = 2563, the size needed to see differences like the paper's.

8.2 Number of bins

Method	5 bins	10 bins	15 bins	20 bins	10 bins, equal counts
CoT	12.35	13.35	13.35	13.72	14.92
Chameleon	8.08	8.08	9.68	8.59	6.19
Chameleon+	5.62	5.62	5.62	5.62	5.62
vPGM-N3	6.61	9.99	11.39	10.96	7.20
vPGM-N4	13.86	13.86	13.86	13.86	9.29

Table 4. ECE (×10²) against the binning choice.

ECE depends on a choice with no right answer. vPGM-N3 goes from 6.6 (5 bins) to 11.4 (15 bins), and with 5 bins it is even lower than Chameleon (8.1), the opposite order from 10 bins. vPGM-N4 falls from 13.9 to 9.3 with equal-count bins. The ECE ranking can change with a setting, so ECE must be read together with the other measures.

8.3 Missing data, noise and graph density

Noisy or missing evidence is part of the evidence experiment of the extension (Section 10). Graph density does not apply here, because the vPGM structures are fixed by the official prompts. The effect of the number of hidden variables is in Section 9.1.

8.4 Effect of the runner bugs

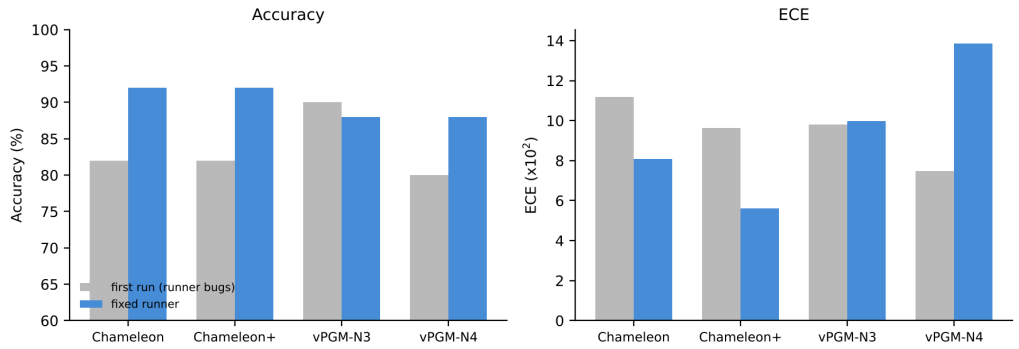


Figure 5. Accuracy and ECE, first runner (grey) against fixed runner (blue).

Method	Correct /50, first	ECE ×10², first	Correct /50, fixed	ECE ×10², fixed
Chameleon	41	11.19	46	8.08
Chameleon+	41	9.64	46	5.62
vPGM-N3	45	9.79	44	9.99
vPGM-N4	40	7.47	44	13.86

Table 5. The same 50 questions with the first and the fixed runner.

This was the biggest effect in the project. The fix raised Chameleon and Chameleon+ from 82% to 92% and vPGM-N4 from 80% to 88%, and it changed vPGM-N4's ECE from 7.47 to 13.86, so the first N4 looked far better calibrated than it is. The first N4 had 7 questions that fell back to a uniform guess; four of them were right only because the tie-break picks option A. In the fixed run only 2 questions have no readable answer list (4918 and 5211), and 146 of 150 answer lists are complete.

- **Evidence from the saved first-run files.** In the first vPGM-N4 file none of the 150 responses reaches the end marker `</EOR>`, and 90 of 150 answer lists are incomplete; many stop in the middle of the list (for example after "P(Y=D | Q,C,M,I,K)=0.025 </"). In the first vPGM-N2 file the word "Hematite", from the last in-context example, appears in all 150 responses, including those for unrelated questions, and in none of the 150-response files of the other methods.
- **vPGM-N2.** Its first run was invalid, not just weak: all 50 prompts were cut by 440 to 881 tokens, the model continued the last in-context example, and the parser returned uniform 0.25 answers. I do not report any N2 number. A valid N2 needs a full re-run with its long prompts (up to 2929 tokens), which did not fit in this milestone.

Lesson. The differences between methods in Table 1 (at most 6 points) are smaller than the changes caused by these bugs. A small comparison is only as reliable as the code that runs it, so every result file is now checked for cut prompts, cut answers and fallback answers before it is used.

9. Ablation: What Each Part Does

9.1 Number of hidden variables

vPGM-N3 and N4 are right on different questions in only 2 of 50 (one each, McNemar p = 1.00), and their Brier scores are close (0.171 and 0.146, Wilcoxon p = 0.097). The extra variable in N4 (Z4, whether the question is ambiguous) is almost always 0.10, so it adds very little (Section 9.3). The N = 2 point of the comparison is missing because of the invalid first N2 run.

9.2 BayesVPGM step: votes against vPGM probabilities

The repository has the posterior model and the loss classes, but no script that fits λ on saved responses, so I wrote my own. It uses the posterior of Section 3 with p_k as the mean of the normalised per-sample probabilities and n_k as the number of samples whose top option is k (my reading of the paper). λ is picked from 61 values between 0.01 and 10,000 by minimising cross-entropy plus the class-wise calibration term (β = 1, the repository default). To avoid fitting and testing on the same questions I used 5-fold cross-validation repeated 10 times: λ is fitted on 40 questions and used on the other 10. I also compared the two parts alone: votes only (λ = 0) and vPGM probabilities only (very large λ).

vPGM-N3, 50 questions	Acc. %	ECE ×10²	Brier	Log-loss
Stored vPGM answer (no λ step)	88	9.99	0.171	0.336
Votes only (λ = 0)	84	11.67	0.219	0.825
vPGM probabilities only	88	8.30	0.165	0.329
λ fitted on all 50 (λ = 31.6), reference only	88	9.91	0.167	0.328
BayesVPGM, cross-validated (mean ± SD, 10 repeats)	87.6	10.07 ± 2.33	0.173 ± 0.003	0.339 ± 0.005
same, β = 0	88	9.93 ± 2.31	0.173 ± 0.003	0.341 ± 0.007

vPGM-N4, 49 valid questions	Acc. %	ECE ×10²	Brier	Log-loss
Stored vPGM answer (no λ step)	88	13.86	0.146	0.293
Votes only (λ = 0)	86	10.00	0.178	0.500
vPGM probabilities only	86	13.97	0.154	0.304
λ fitted on all (λ = 4.0), reference only	84	11.81	0.153	0.280
BayesVPGM, cross-validated (mean ± SD, 10 repeats)	84	8.82 ± 1.32	0.171 ± 0.006	0.331 ± 0.054
same, β = 0	84	8.30 ± 1.05	0.169 ± 0.005	0.322 ± 0.053

Table 6. Parts of the BayesVPGM step. For N4 one question has no readable answer list and is left out of the fitted rows. The stored answer is the one used in Table 1.

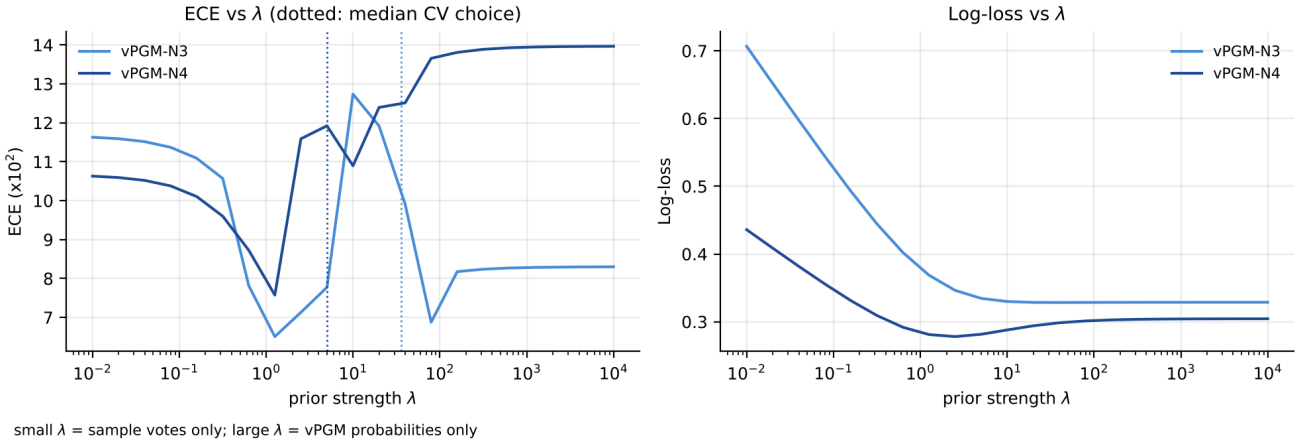


Figure 6. ECE and log-loss against λ.

What I see.

- The three votes alone are a poor probability: log-loss 0.82 (N3) and 0.50 (N4), against 0.33 and 0.30 for the vPGM probabilities. Most of the information is in the vPGM probabilities.
- The fitted mix is about as good as the probabilities alone (log-loss 0.339 against 0.329 for N3), and switching off the calibration term (β = 0) changes almost nothing. On this data I cannot show that the calibration loss helps.

Does the step improve calibration? Not consistently.

- N3: the cross-validated ECE (10.07 ± 2.33) equals the value without the step (9.99), and 17 of the 50 fits chose the largest λ, which means they threw the votes away.
- N4: ECE falls from 13.86 to 8.82 ± 1.32, but the Brier score gets worse (0.146 to 0.171), the log-loss gets worse (0.293 to 0.331) and accuracy drops from 88% to 84%. The chosen λ jumps from 0.01 to 398.
- I think the ECE fell because the model became less confident, not because it became better calibrated. Figure 6 supports this: ECE jumps with λ (answers move between bins) while log-loss is smooth and flat above λ ≈ 10. Choosing λ by ECE is unstable on 50 questions.
- The paper's 1.05 and 1.50 are not reproduced and are also not testable here, since the noise floor is about 9 to 10.
- My version differs from the paper's: no separate validation set, a grid search instead of L-BFGS, and my own reading of p_k and n_k. A fair test needs a few hundred questions.

9.3 Do the hidden variables matter?

Average stated value over 3 answers	N3 right	N3 wrong	N3 p	N4 right	N4 wrong	N4 p
Z1: outside knowledge needed	0.53	0.62	0.69	0.43	0.52	0.44
Z2: caption reliable	0.36	0.16	0.20	0.33	0.24	0.59
Z3: detected text reliable	0.50	0.50	–	0.50	0.50	–
Z4: question ambiguous (N4 only)	–	–	–	0.10	0.10	0.47
Share of 3 answers agreeing with the majority	0.97	0.83	–	0.99	0.72	–

Table 7. Hidden variables for right and wrong answers (6 wrong each for N3 and N4). p is from the Mann-Whitney test; a dash means no spread or no test.

None of the hidden variables separates right from wrong answers (all p above 0.19, with only 6 wrong answers). Z3 is exactly 0.50 on all 50 questions, because no question in this subset has detected text, and Z4 is about 0.10 everywhere. Z2 is lower for wrong answers (0.16 against 0.36 for N3) but this is not significant. Agreement between the three answers does carry a clear signal: when all three agree (44 questions for N3) the answer is right 93% of the time, and when they do not, only 50% (6 questions; for N4, 20% of 5).

What this tells me. The better ranking of mistakes in Section 7 does not come from Z1 to Z4 on this data. It comes from the final answer probabilities and from the disagreement between the three answers. This is a limit of the test: most of these questions have no useful caption or detected text, so Z2 and Z3 have little to judge. That is the reason for my extension. A reliability variable can only show its value on questions that come with evidence of different quality, so the evidence experiment has to build such evidence on purpose.

10. The Extension: Reliability-Weighted Update and ΔC

What is done. The update rule of M2 is implemented as code together with the base posterior. The code reproduces the worked example of M2 and checks the proved properties:

	Class 1	Class 2	Class 3
π before (p = (0.6, 0.3, 0.1), λ = 5, votes (2, 1, 0))	0.6250	0.3125	0.0625
π after evidence for class 2 (r = 0.5, κ = 1, $\bar{s}$ = (0, 1, 0))	0.5882	0.3529	0.0588
Change	−0.0368	+0.0404	−0.0037

The checks confirm that the update equals the convex mix $(1 - w)\pi + w \cdot \bar{s}$, and that r = 0 leaves the belief unchanged.

What is not done. The update has been tested only on given values of r and s̄. It has not been run on reliabilities and answer distributions produced by the LLM, so the four ΔC predictions of M2 are still untested. The experiment is designed and fixed before any result: for about 40 of the 50 questions the starting belief is the saved vPGM-N3 answer, each question gets evidence of five kinds (reliable support, irrelevant, weakly misleading, strongly misleading, and two conflicting items), the LLM gives r and s̄ for each item, and the update is applied offline with a version that ignores reliability (r = 1) as the comparison. ΔC is measured on the answer that was most likely before the evidence, with Wilcoxon tests and resampled intervals. This is the main work of the final milestone. I do not claim any ΔC result in this report.

11. Error Analysis

Method	Wrong	Question ids answered wrongly (confidence)
CoT	7	4483 (0.95), 4568 (0.95), 4832 (0.95), 4836 (0.90), 4918 (0.25), 5018 (0.85), 5052 (0.95)
Chameleon	4	4836 (0.95), 4883 (0.95), 5018 (0.95), 5052 (0.85)
Chameleon+	4	4832 (1.00), 4836 (0.52), 5018 (0.50), 5052 (1.00)
vPGM-N3	6	4836 (0.68), 4883 (0.87), 4918 (0.30), 5018 (0.78), 5211 (0.35), 5238 (0.57)
vPGM-N4	6	4568 (0.57), 4836 (0.74), 4883 (0.57), 4918 (0.25), 5018 (0.42), 5238 (0.60)

Table 8. Wrong answers of each method.

- Ten of the 50 questions are missed by at least one method, eight by two or more, and two (4836 and 5018) by all five.
- On these two, CoT (0.90 and 0.85) and Chameleon (0.95 and 0.95) are very sure, while the others are less sure: Chameleon+ 0.52 and 0.50, vPGM-N3 0.68 and 0.78, vPGM-N4 0.74 and 0.42. This is the pattern of Section 7: the baselines fail with high confidence and vPGM is more careful.
- Both vPGM variants also miss 4883, 4918 and 5238. Chameleon misses only 4883 and Chameleon+ none, so these are vPGM's extra errors.
- By number of options: with three options (21 questions) every method gets 3 wrong. With four options (29 questions) CoT gets 4 wrong, Chameleon and Chameleon+ get 1, and vPGM-N3 and N4 get 3. The extra vPGM errors are on the harder four-option questions.
- **A limit of my pipeline, not a reasoning error:** question 4918 is a map of Caribbean countries and cannot be answered without the image, but my pipeline passes only text. CoT said it is impossible to determine the country and picked A with confidence 0.25, and the vPGM answers have confidence 0.25 to 0.30. These are honest "I do not know" answers with correctly low confidence, but they still count as wrong. I kept them because all methods were tested on the same questions.
- **No-answer cases:** in the fixed runs a few responses contain no usable answer (for example "none of the above" or "[OPTION]"). The original code replaces these with a random guess, so I kept that rule. For CoT and Chameleon these cases have a low confidence (0.25 to 0.50), so they do not add overconfidence. Chameleon+ is different. Its confidence is mostly the share of its three answers that pick the same option and a random guess can count as a vote. Stored confidence usually equals the vote share, with exceptions (pids 4720, 5119, 4883, 5018, 4836). So question 5052 is wrong with confidence 1.00, although two of its three answers were random guesses with a stated probability of 0.333.

12. Revised SOTA Positioning

At M1 I placed this project on top of BayesAgent [1], still the closest work. Following Identify → Understand → Compare → Validate → Reposition:

1. **Identify.** The paper claims that vPGM and BayesVPGM keep accuracy and cut ECE on ScienceQA from about 9 to about 1 or 2.
2. **Understand.** Hidden variables stated in words by the LLM are combined with the votes of several answers through a Dirichlet prior, whose strength is fitted with a calibration loss.
3. **Compare.** On 50 questions with the same model, accuracy is consistent with the paper. The ECE result is not reproduced, and my test is too small to resolve it. What my data does support is that vPGM ranks its right and wrong answers better (AUROC 0.94 and 0.95 against 0.70 to 0.80), and that both variants have a better log-loss than CoT and Chameleon+.
4. **Validate.** My cross-validated BayesVPGM step gives no consistent gain. The hidden variables Z1 to Z4 do not separate right from wrong answers on this subset. And the implementation details (input cut, output cap, parser fallbacks) changed the results by more than the methods differ from each other.
5. **Reposition.** The project is now narrower than I said at M1. First, it is a small and carefully checked reproduction of Table 1 of [1]: accuracy matches, vPGM's better ranking of its mistakes is supported against CoT and suggestive against Chameleon, and the ECE claim cannot be tested at this size. Second, the new part is my extension: a reliability node for each evidence item, an update rule with proved properties (Section 6 of M2) that now exists as checked code, and the ΔC measurement, which is still to be done. One caution: judging the quality of a source is not new in itself, because Z2 and Z3 already do it for the caption and the detected text. What is new is the general form with any number of evidence items and the proved update.

13. Limitations

- **Sample size.** With 50 questions an accuracy interval is about ±9 points, and the ECE of vPGM is inside what luck alone would give. Every comparison is preliminary.
- **One run per method.** The milestone asks for several seeds, but each vPGM run takes about 14 hours or more on this server (about 17 minutes per question for N4, and about 26 for N3). My uncertainty comes from resampling the questions, not from running the model again, so run-to-run variation of the LLM is not measured. CoT and Chameleon are cheap enough to repeat, and that is the first thing I would add.
- **Sample-matched comparison.** CoT and Chameleon use one answer and vPGM uses three. Chameleon+ is the fair baseline for vPGM, and against it I see no clear difference in accuracy or ECE.
- **Brier score and log-loss for CoT and Chameleon** depend on how I spread the remaining probability. I split it equally.
- **Intervals and noise floor** are good indications, not exact tests. Resampled ECE is biased upward on small samples, and the noise floor assumes that the stated confidences are the true ones.
- **The λ step** is my own version, without a validation set, with a grid search instead of L-BFGS and my own definition of the votes. The hidden-variable test has only 6 wrong answers per method.

- **Text-only pipeline.** Some questions need the image (for example 4918), and the prompt has no detected-text field, so Z3 has nothing to judge.
- vPGM-N2 has no valid result, and the extension has not yet been run on an LLM.
- **Not done:** A-OKVQA clean and noisy, ChatCoach, other LLMs, a direct comparison of the transformers backend with vLLM, and ANOVA across methods.

14. Reproducibility

The M3 folder follows the course layout: M3_<group_name>/ with README.md, Report/, Code/src/ (with its own README.md), Results/ and Data/README.md. The code is the official repository of [1] with my transformers runner (no input cut, a stop on `</EOR>`, fixed seeds), the BayesVPGM cross-validation script, the extension code and the analysis scripts of this milestone. The analysis needs only the saved result files and runs on a CPU in a few minutes. Three scripts produce everything in this report: one for the metrics, tests, noise floor and sample-size curves; one for the BayesVPGM step, hidden variables, error analysis and the before-and-after comparison of the runner fix; and one for all figures.

Seeds. Resampling and simulations use seed 2026, and the cross-validation repeats use seeds 0 to 9. The model runs used seed 0, temperature 0.5, top-p 0.95, 600 new tokens and the stop marker. The result files are in results/my_scienceqa/test_50/: CoT run-0, and run-1 for Chameleon, Chameleon+, vPGM-N3 and vPGM-N4 (fixed runner). The older files are kept only for the comparison in Section 8.4.

15. Timeline and Next Steps

Phase	Planned work	Status after M3
Phases 1 and 2	Understand the problem, reproduce BayesAgent and the baselines	Done at n = 50 with the fixed runner (N2 excluded)
Phase 3 (Weeks 7-9)	Multiple evidence and controlled evidence conditions	Design fixed; update rule coded and checked
Phase 4 (Weeks 10-12)	Sequential evidence and Bayesian belief updates	Planned for the final milestone
Phase 5 (Weeks 13-14)	Ablation, calibration analysis, final comparison	Done for N3 and N4; N2 missing
Phase 6 (Final)	Final paper and presentation	Planned

Plan for the final milestone, in order of value:

1. Run the evidence experiment with the LLM and test the four ΔC predictions.
2. Re-run vPGM-N2 with the fixed runner and complete the hidden-variable comparison.
3. Repeat CoT and Chameleon with several seeds.
4. Run a few hundred questions with a newer driver or an API, the only way to test the ECE claim, and fit λ on a separate validation set.

16. Conclusion

M3 asked what the evidence says. The main finding came from checking my own runner: three bugs changed the results by more than the methods differ from each other, and they made the first vPGM-N2 and vPGM-N4 results meaningless. With the corrected runner, the five valid methods cannot be separated by accuracy on 50 questions (86 to 92%), and the numbers are in the range of the paper. The paper’s ECE advantage for vPGM was not reproduced, but it also could not be tested, because the ECE of vPGM is about what a perfectly calibrated model would

show by luck at this size. What vPGM does better is ranking its own mistakes (AUROC 0.94 to 0.95) and log-loss. It is not better in ECE or on the typical question. The BayesVPGM step gave no consistent gain, and the hidden variables do not separate right from wrong answers on this subset. The update rule of my extension is implemented and checked, and its ΔC behaviour on LLM-produced evidence is the work of the final milestone.

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